The Probability of a Lag Spike: Statistical Solution for Wi-Fi Latency

Moving Beyond "Average Ping" Using Right-Skewed Probability Distributions, Goodness-of-Fit Testing, and Tail Risk Inference

Presenter

Saksham [Applied Probability & Inference]

Course & Implementation Scope

Applied Probability & Statistical Inference
Empirical Wi-Fi Dataset: N = 3,685 ICMP Echo Probes
Evaluation of Right-Skewed Models & Lag Spike Tail Risk

The Problem: Why "Average Latency" Lies to You on Wi-Fi

The mean latency appears moderate, but tail volatility and severe skewness create disruptive lag spikes.

WHAT PEOPLE ASSUME

Symmetric Normal Distribution

- **Bell-Curve Assumption:** Belief that latency symmetrically clusters around the 176.8 ms average.
- **Predicts Negative Latency:** Normal model predicts a 13.1% probability of RTT < 0 ms (physically absurd!).
- **Severe Tail Blindness:** Estimates probability of catastrophic lag (>800 ms) as 0.003% (virtually zero).
- **Misleading SLA Audits:** ISPs report mean ping, masking user disconnections and frame freezes.

EMPIRICAL WI-FI REALITY

Right-Skewed & Heavy-Tailed

- **Strict Propagation Floor:** Physical minimum latency bound at 17.9 ms (no values below this).
- **Extreme Right-Skewness:** Skewness = +3.560 (heavily asymmetrical long right tail).
- **Leptokurtic Kurtosis:** Excess Kurtosis = +26.00 (extreme outlier probability).
- **Catastrophic Latency Spikes:** Max observed ping = 2479.0 ms (14× higher than the 176.8 ms mean!).
- **High Disruption Rate:** P(X > 100 ms) = 64.0% (64% of packets experience lag!).

TAKEAWAY **Mean latency (176.8 ms) masks an extreme 2479 ms tail spike; right-skewed models are mathematically mandatory.**

Exploratory Data Analysis: Wi-Fi Empirical Characteristics

Rigorous distributional summary of N = 3,685 consecutive ICMP ping packets collected over Wi-Fi.

Central Tendency vs Spread

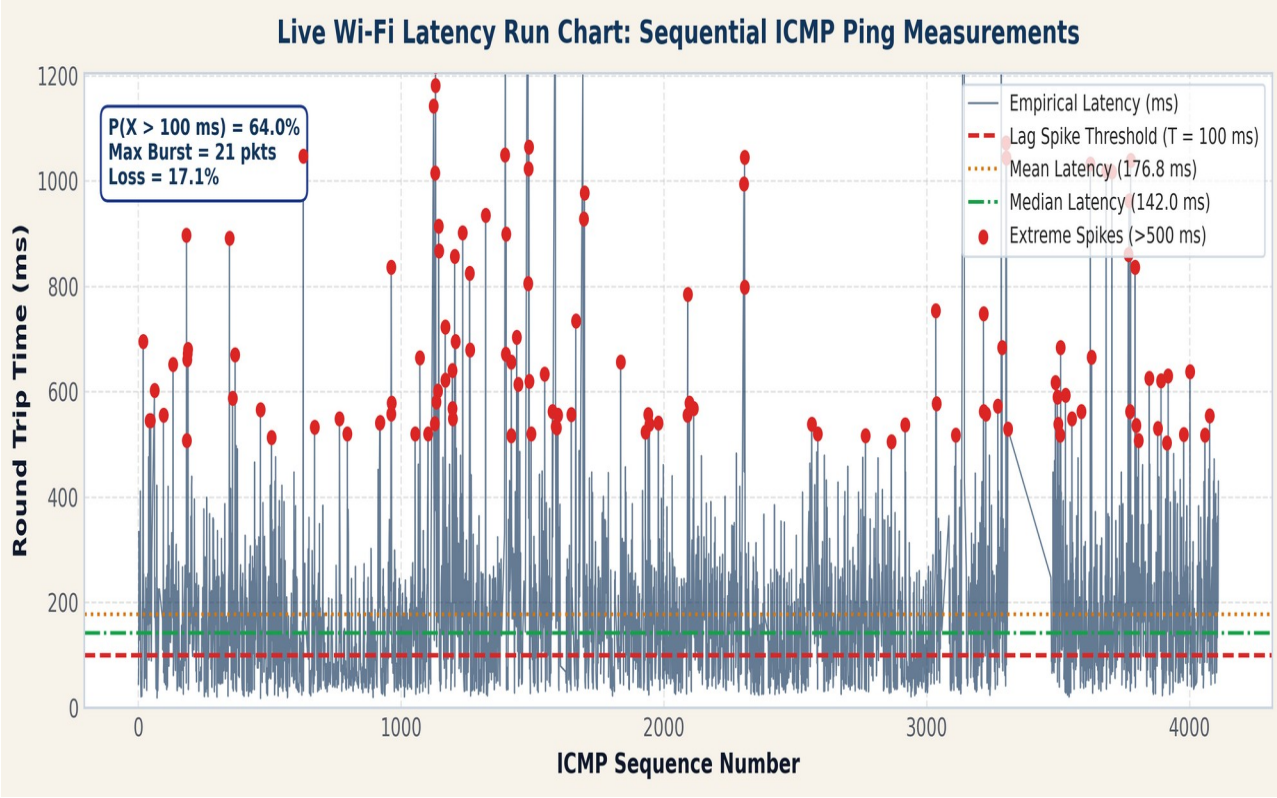
- Mean: 176.81 ms | Median: 142.00 ms
- Std Dev: 157.99 ms | IQR: 159.10 ms
- Minimum: 17.90 ms | Maximum: 2479.00 ms

Percentile Spectrum

- 50th Percentile (P50): 142.00 ms
- 90th Percentile (P90): 340.00 ms
- 95th Percentile (P95): 444.60 ms
- 99th Percentile (P99): 826.76 ms (Severe Tail)

Higher Moments & Network Reliability

- Skewness: +3.560 (Severe Right Asymmetry)
- Excess Kurtosis: +26.00 (Fat Tails)
- Packet Loss: 17.08% (759 / 4,444 packets dropped)



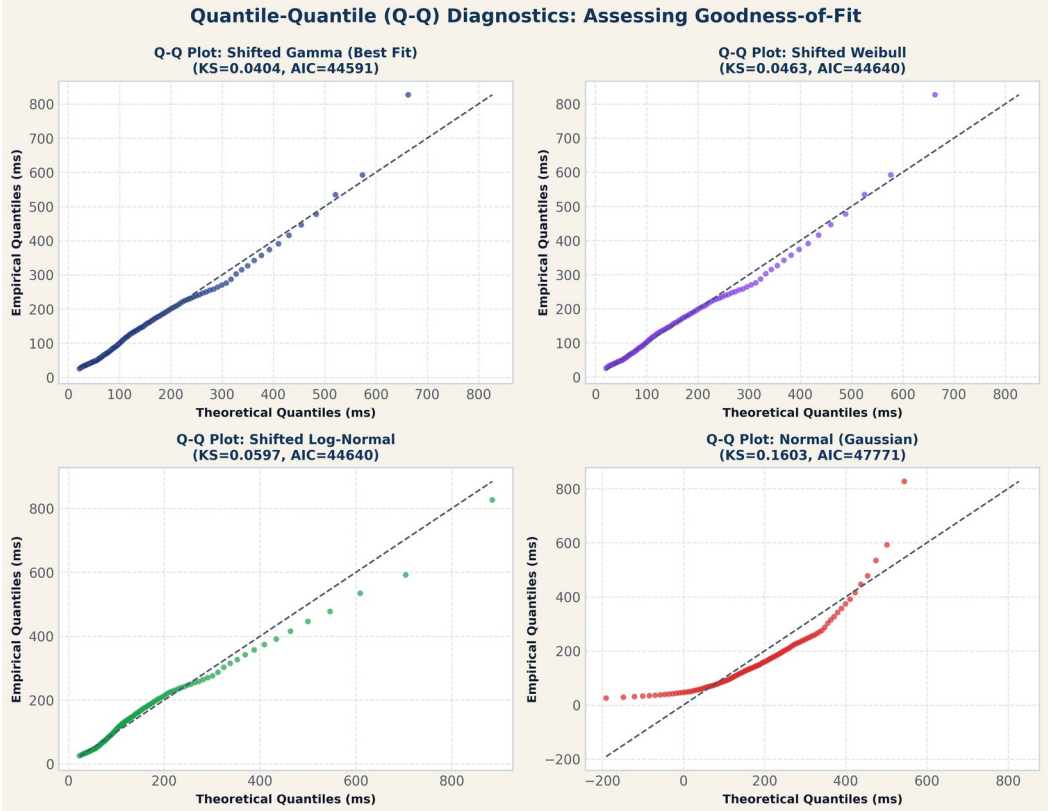
TAKEAWAY **Median latency (142.0 ms) is 35 ms below the mean; 1% of packets (P99) exceed 827 ms with 17.1% packet loss.**

Hypothesis Testing: Statistical Rejection of Gaussian Normality

Multiple formal statistical tests decisively reject the null hypothesis of normal latency (p < 1e-82).

Formal Goodness-of-Fit Tests

- **Kolmogorov-Smirnov (K-S) Test:**
  D = 0.1603, p = 3.53e-83
  Decisive rejection of Gaussian CDF at alpha = 0.001.
- **D'Agostino-Pearson Omnibus Test:**
  K^2 = 2967.5, p = 0.000
  Simultaneously tests skewness and kurtosis components.
- **Excess Kurtosis Test:**
  Excess kurtosis = +26.00 (Normal = 0)
  Extreme leptokurtic peakedness and heavy right tail.
- **Conclusion:**
  The empirical data is fundamentally non-Gaussian; parametric modeling requires strictly positive, skewed distributions.



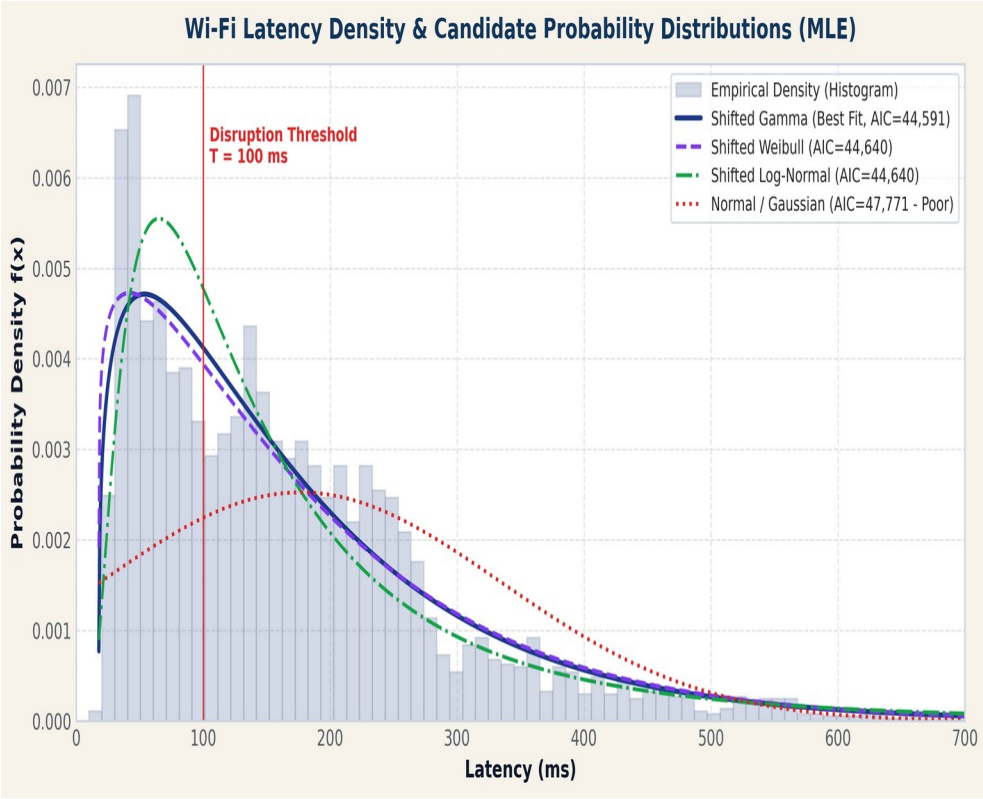
TAKEAWAY **Normality is rejected with p < 1e-82; Q-Q diagnostics demonstrate that Gaussian models deviate catastrophically in the upper tail.**

Distribution Modeling: Maximum Likelihood Estimation (MLE)

Evaluating right-skewed parametric families incorporating the physical propagation floor (gamma = 17.9 ms).

Distribution Model	Log-Lik	AIC	BIC	K-S Stat
Shifted Gamma (Best)	**-22,292.4**	**44,590.9**	**44,609.5**	**0.0404**
Shifted Weibull	-22,317.1	44,640.3	44,658.9	0.0463
Shifted Log-Normal	-22,317.2	44,640.3	44,659.0	0.0597
Log-Normal (loc=0)	-22,322.1	44,648.2	44,660.6	0.0500
Shifted Exponential	-22,361.8	44,727.5	44,739.9	0.0588
Normal (Gaussian)	-23,883.7	47,771.4	47,783.8	0.1603

Best Model Parameters: 3-Parameter Shifted Gamma
- Shape (alpha) = 1.2927 | Scale (beta) = 122.95 ms | Loc (gamma) = 17.87 ms
- Physical interpretation: gamma represents the hardware propagation floor.



TAKEAWAY **Shifted Gamma achieves the lowest AIC (44,591) and K-S statistic (0.0404), outperforming Weibull, Log-Normal, and Normal.**

Beyond the Mean: Calculating the Probability of Disruption

Evaluating lag spike disruption risk: P(X > 100 ms) = 1 - F_X(100), comparing empirical reality to proposal estimates.

Threshold Disruption Analysis

T = 100 ms Disruption Evaluation:

- Proposal Illustrative Placeholder: ~4.5% (Initial Wi-Fi assumption)
- Empirical Live Wi-Fi Reality: 64.04% (2,360 / 3,685 pkts)
- Shifted Gamma Fitted Model: 64.34% (Relative error < 0.5%)
- Shifted Weibull Model: 63.72% | Log-Normal: 61.96%

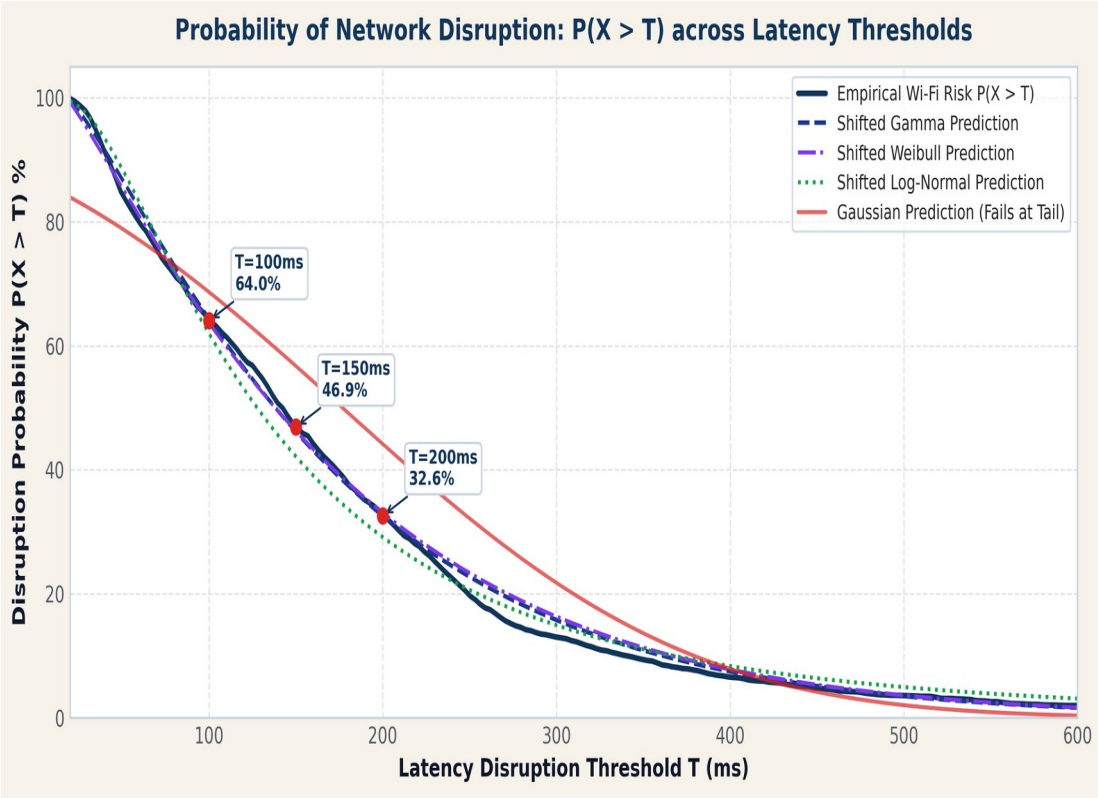
T = 150 ms & 200 ms Thresholds:

- P(X > 150 ms): Empirical = 46.89% | Shifted Gamma = 46.15%
- P(X > 200 ms): Empirical = 32.59% | Shifted Gamma = 32.57%

Proposal Placeholder (Wi-Fi) (4.5%)

Empirical Live Capture (Wi-Fi) (64.0%)

Shifted Gamma MLE (Wi-Fi) (64.3%)



TAKEAWAY **Empirical Wi-Fi disruption at 100 ms is 64.0% (vs 4.5% proposal estimate); Shifted Gamma predicts 64.34% with remarkable precision.**

Uncertainty Estimation: 95% Bootstrap Confidence Intervals

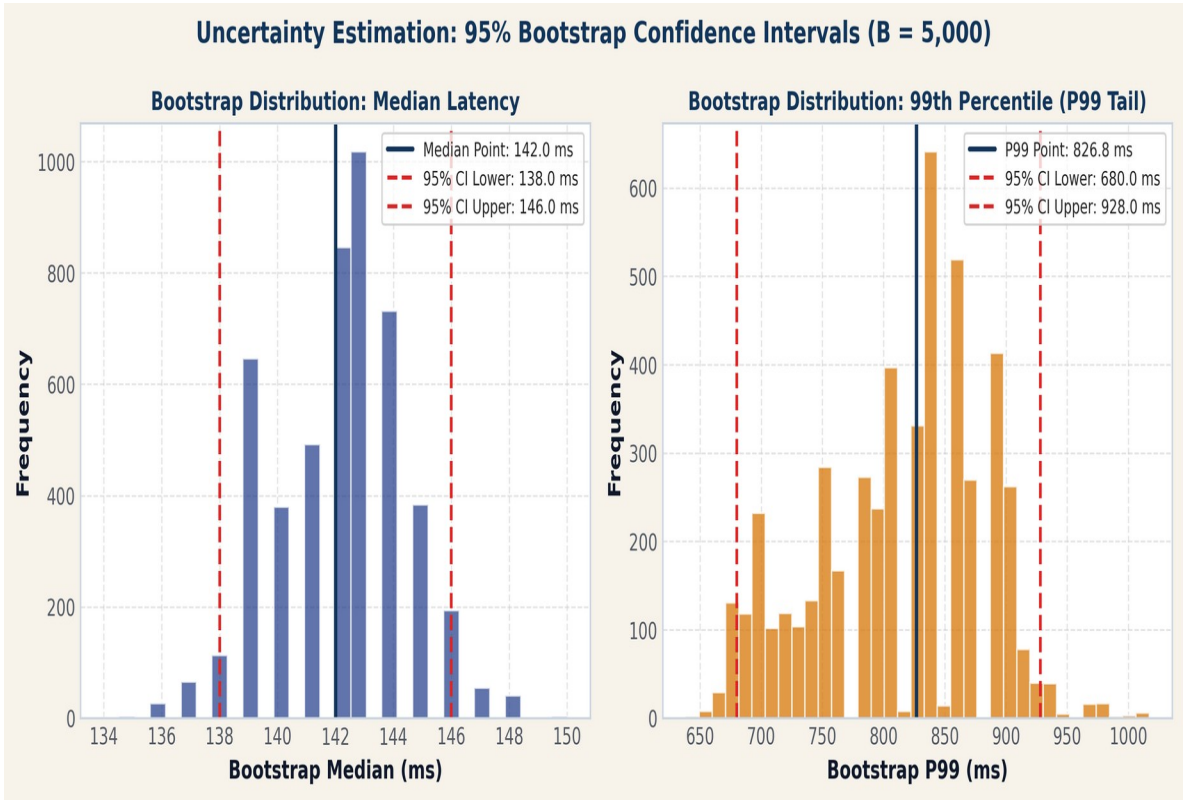
Quantifying parameter stability via B = 5,000 non-parametric bootstrap resamples on live Wi-Fi latency observations.

Median Latency [P50]: 142.00 ms **95% CI: [138.00 ms, 146.00 ms]**
Extremely tight bound (+-4 ms); represents typical packet latency.

Mean Latency: 176.81 ms **95% CI: [171.87 ms, 182.00 ms]**
Influenced by right-tail outliers; spans +-5.1 ms.

90th Percentile [P90]: 340.00 ms **95% CI: [325.00 ms, 357.00 ms]**
Substantial tail delay (+-16 ms); critical for interactive cloud gaming.

99th Percentile [P99 Tail]: 826.76 ms **95% CI: [680.00 ms, 928.00 ms]**
High tail volatility spanning [680 ms, 928 ms]; causes frame freezes.



TAKEAWAY **Non-parametric bootstrap confirms median stability [138-146 ms] alongside extreme tail volatility [680-928 ms].**

Temporal Dynamics, Burstiness & Network Engineering Conclusions

Wi-Fi lag spikes are not independent; temporal clustering exacerbates user disruption and demands modernized SLAs.

1 · Temporal Dynamics

- **Lag-1 Autocorrelation:**
  rho_1 = +0.270 (p < 1e-15).
Consecutive packets share network queuing states.
- **Disruption Clustering:**
  642 distinct spike bursts (>100 ms).
Mean burst length = 3.7 packets.
- **Extreme Burst Duration:**
  Maximum burst = 21 consecutive packets.
Translates to ~21 seconds of continuous degradation!
- **Engineering Reality:**
  Spikes occur in clustered waves rather than memoryless Poisson arrivals.

2 · Disruption Impact

- **Video Conferencing (Zoom/Teams):**
  100 ms+ jitter bursts exceed playout buffers, causing frozen frames and clipped audio.
- **Esports & Cloud Gaming:**
  P90 (340 ms) and P99 (827 ms) trigger server desync, input drops, and missed tick-rates.
- **Financial Transactions:**
  Deterministic latency bounds are violated; time-sensitive executions risk slippage.
- **Composite Quality Score:**
  17.1% packet loss + 64.0% lag spikes = only 29.8% of packets are on-time and healthy.

3 · Engineering Solutions

- **Redefine ISP SLAs:**
  Replace deceptive 'average ping' guarantees with Shifted Gamma tail percentiles (P95, P99) and P(X > 100 ms).
- **Active Queue Management (AQM):**
  Implement FQ-CoDel or Cake AQM on Wi-Fi access points to eliminate bufferbloat queuing spikes.
- **Adaptive Jitter Buffers:**
  Real-time communication software must dynamically size jitter buffers up to ~850 ms during Wi-Fi bursts.
- **Project Methodology Success:**
  Empirical analysis validates the 4-step framework proposed in Applied Probability course.

TAKEAWAY **Wi-Fi latency spikes exhibit temporal autocorrelation (rho_1 = 0.27); network engineering must replace average ping with Shifted Gamma tail metrics.**